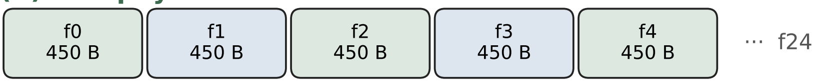
VUS / USEIS cassette layout (Viking Lander 2)

Ground-processed frames — dual-archive ground truth (UTIG vkg.47–56)

(A) UTIG cassette file (one subgroup)

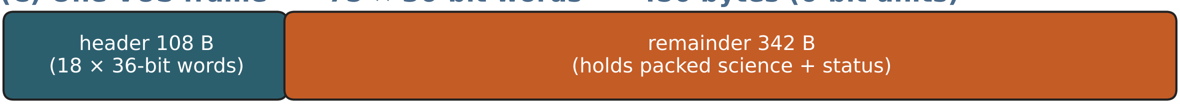
Header tape label ASCII “VUS…” · record_length = 11250

(B) One physical record = 25 consecutive VUS frames



Frame index advances by 450 B within the concatenated record body.

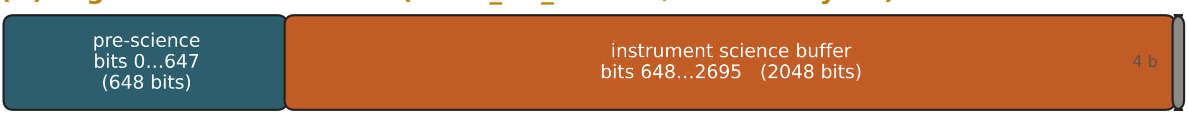
(C) One VUS frame — 75 × 36-bit words = 450 bytes (6-bit units)



On media: each 36-bit word = six 6-bit cells (one byte each, low 6 bits used).

Halfword view of header: 36 × 18-bit halfwords — same eng. fields as SEISF header.

(D) Logical 2700-bit stream (make_bit_stream / vusinfo layout)



Science offset = (word 19 − 1) × 36 = 648. Path D injects recovered SEISF bits into this region.

Command / mode / GCSC and amplitude blocks are parsed from this stream (vusinfo).

(E) Science buffer pages (instrument DRAM readout order inverted on ground)

On VUS this 2048-bit region is already sequential. On SEISF it is scrambled until Path D.

Validation role

Header-matched pairs (same year/DOY/GCSC) let Path D assert bit identity: 2048/2048 residual 0 and 450-byte frame equality on gold cases (e.g. GCSC = 125078, year/DOY 1976/249).